

Certified Robustness for Model-Free PID Tuning by Step-Response Inspection

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Abstract

The SPIN method tunes a PID controller without a model. After each setpoint step it counts how many turns the control error makes in three phase portraits, and it raises the gains until one count is above its limit. The limits were chosen by calibration, and nothing guaranteed the robustness of the result. Our results are twofold. First, for tunings that cancel the plant lags, the counts have an exact meaning. The loop has one parameter k , its gain margin is $\pi/(2k)$, and the error crosses the setpoint only if $M_s > 1.39$. Each crossing adds exactly half a turn to the count, so at the default settling band of 2% a count of at most half a turn guarantees $M_s \leq 1.53$, and each calibrated limit of SPIN turns out to separate n from $n + 1$ crossings. Second, for any stable loop, the counts and M_s are not tied in either direction, but the error after a setpoint step is the step response of the sensitivity function. So the step itself gives M_s , and the loop $L = 1/S - 1$, so it also predicts the peak of any other tuning of the same plant. We use both as an acceptance test and a shaping rule inside SPIN: if the peak is too high, SPIN lowers the band whose move is predicted to help, and we prove when such a band exists. On 19 test loops and two start tunings, SPIN as published ended above $M_s = 1.7$ on 8 or 9 loops, up to 2.07, depending on the start. With the rule, all 19 loops ended at or below the target from both starts, within a reading error below 1%, at a mean load-rejection cost of 5 to 8%. An ablation shows that the rule, not the gain box, does this.

1 Introduction

Let $L = CG$ be the loop transfer function and $S = 1/(1 + L)$ the sensitivity function. The peak $M_s = \max_{\omega} |S(j\omega)|$ measures robustness. A value near 1.4 is safe, and a value near 2 is close to the limit [2]. Tuning rules reach a chosen M_s by using a model [2, 3]. After tuning, plants rarely check M_s , be-

cause a check needs a model or a test that disturbs production [5, 6].

1.1 The SPIN method

SPIN [1] tunes a PI or PID controller $C(s) = K_p + K_i/s + K_d s$ without a model. In each iteration it applies one setpoint step and records the control error e . From this record it forms three phase portraits,

$$\Gamma_0 : (e, z), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \quad (1)$$

where $z(t) = \int_0^t e dt'$ is the running integral of e and Δ is the difference between samples. Each portrait reacts to one frequency band. The bands belong to K_i , K_p and K_d . The turn count N_k is the number of turns that Γ_k makes around its end point before the error stays within a band δ (by default 2% of the step). Each count has a limit $\bar{N}_k$, and Algorithm 1 gives the rule. An oscillation shows in its own band and in all higher bands, so the lowest band above its limit is the one that causes it. SPIN needs no model, no identification and no plant-specific setting: the same dimensionless defaults served every plant in [1]. But the limits were chosen by calibration, and nothing guaranteed the robustness of the result. This paper adds one test to line 7 (Algorithm 2 in Section 4). An online simulator lets the reader try SPIN.¹

1.2 Two results

Result 1, exact, for lag-cancelled loops (Section 2): each setpoint crossing adds exactly half a turn to a count, a limit of $n/2$ turns guarantees a tabulated M_s , and the calibrated limits of SPIN separate the same crossing numbers. The tools (Lambert W , the bound $k \leq 1/e$) are classical [10, 11]; their link to the counts is new. *Result 2, for any stable loop* (Sections 3, 4): the known identity $S = sE$ becomes an acceptance test and a shaping rule inside a model-free tuner. Both are theory checked in simulation; noise and real loops are left to a follow-up.

1.3 Related work

A frequency response can be computed from step or relay responses by the Fourier transform [7]. The sensitivity func-

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¹<https://robopid-simulator.uceeb.cvut.cz>

Algorithm 1: one iteration of SPIN [1]

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1: Apply a setpoint step and record the error  $e$ .
2: if the record does not settle then
3:   divide  $K_i, K_p, K_d$  by 2, 4, 8 (back off)
4: else
5:   count the turns  $N_0, N_1, N_2$  of  $\Gamma_0, \Gamma_1, \Gamma_2$ 
6:   if  $N_k \leq \bar{N}_k$  for every band  $k$  then
7:     raise all gains by one step (accept)
8:   else
9:      $k \leftarrow$  lowest band with  $N_k > \bar{N}_k$ 
10:    lower the gain of band  $k$  by one step,
    raise the gains of the bands below  $k$  by one step
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One step multiplies or divides a gain by $1/(1 - \beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$, each gain within $[0.001, 10]$ times its start. The last accepted tuning is the result.

tion can be estimated from closed-loop data with reference excitation; this is the first stage of the two-stage identification method [8]. The setpoint overshoot method tunes PI controllers from the overshoot and peak time of one closed-loop setpoint step [14]. Margins have been estimated from closed-loop data with designed experiments [9]. Our use is narrower: one normal setpoint step, no special excitation, used inside a model-free tuning rule. Other model-free methods tune by repeated closed-loop experiments, such as iterative feedback tuning [12] and extremum seeking [13], or from one data set, such as virtual reference feedback tuning [15]. They aim at a performance cost; here we certify M_s from data the tuning already produces.

Why not design the controller from the loop L that one step gives? A closed-loop step excites the plant mainly near the present crossover, so a large move must trust L where the data are poor. Certified SPIN uses the prediction only for one small SPIN step, and the next step checks it.

2 Result 1: What a Turn Count Means

2.1 The lag-cancelled loop

Lambda, IMC and SIMC tunings cancel the plant lags [3]. Take a PI controller $K_c(1 + 1/(T_i s))$ with gain K_c and integral time T_i on the plant $Ke^{-sL_d}/(Ts + 1)$ with gain K , lag T and dead time L_d . The choice $T_i = T$ cancels the lag. So does a series PID $K_c(1 + 1/(T_i s))(1 + T_d s)$ with $T_i = T_1$ and derivative time $T_d = T_2$ on a plant with two lags T_1, T_2 . The loop is then an integrator with a dead time,

$$L(s) = \frac{ke^{-sL_d}}{sL_d}, \quad k = \frac{KK_c L_d}{T_i}, \quad (2)$$

with one dimensionless gain k . We call it the lag-cancelled loop.

Proposition 1 (Exact robustness) *The loop (2) is stable for*

$0 < k < \pi/2$. Its gain margin and sensitivity peak are

$$\text{GM} = \frac{\pi}{2k}, \quad \frac{1}{M_s} = \min_{x>0} \frac{\sqrt{x^2 - 2kx \sin x + k^2}}{x}. \quad (3)$$

Its closed-loop poles are $s_j = W_j(-k)/L_d$, where W_j are the branches of the Lambert W function [10].

Proof: The phase of $L(j\omega)$ is $-\pi/2 - \omega L_d$. It reaches $-\pi$ at $\omega L_d = \pi/2$, where $|L| = 2k/\pi$. With $x = \omega L_d$, $|1 + L|^2 = (x^2 - 2kx \sin x + k^2)/x^2$. The poles solve $sL_d e^{sL_d} = -k$. $\square$

SIMC sets $K_c = T/(K(\tau_c + \theta))$ and $T_i = \min(T, 4(\tau_c + \theta))$, where $\theta = L_d$ is the dead time and τ_c the desired closed-loop time constant. With the recommended $\tau_c = \theta$ and $T \leq 8\theta$, it cancels the lag and gives $k = \frac{1}{2}$, so $\text{GM} = \pi$ and $M_s = 1.59$. These are exactly the values in [4, Table 2].

2.2 Why the count grows in half turns

After a unit setpoint step, the error of the loop (2) obeys $\dot{e}(t) = -(k/L_d)e(t - L_d)$.

Proposition 2 (Crossings) *For the loop (2): (a) the error never crosses the setpoint if and only if $k \leq 1/e$, that is $M_s \leq 1.39$. For $1/e < k < \pi/2$: (b) each extremum of the error comes exactly one dead time after a setpoint crossing; (c) from the first undershoot on, each extremum is smaller than the previous one by the factor $q \approx e^{-\pi\alpha/\omega}$, where $-\alpha \pm j\omega$ is the dominant pole pair.*

Proof: (a) The record is the fundamental solution of the delay equation. It stays positive if and only if $k \leq 1/e$ [11]. (b) $\dot{e} = 0$ exactly when $e(t - L_d) = 0$. (c) The dominant pair gives the error; the next pair decays at least 3 times faster. The extrema of a decaying sine are half a period apart. $\square$

Part (c) holds up to the faster poles: for $k \leq 1.4$ the relative error is below 0.3% for the first ratio and 10^{-4} later.

Each setpoint crossing adds exactly half a turn to the portrait: the crossings lie on one line through the end point, on alternate sides (Fig. 1(b,c)). A crossing is counted if the extremum after it is larger than δ . The n -th extremum equals the overshoot times q^{n-1} , and both grow with k (checked numerically for $1/e < k < 1.5$). So the count is a staircase in M_s , with one step per crossing (Fig. 1(a)). Before the first crossing, the curve already turns by about a third of a turn. So a record with n counted crossings has a count strictly between $n/2$ and $(n+1)/2$, and a count at or below $n/2$ means fewer than n crossings. A limit on a half turn therefore lies in the gap between two plateaus of the staircase. There, a small change in the record cannot change the decision. Table 1 gives the M_s that each limit guarantees. The band δ moves these values, but the steps remain half turns. As $\delta \rightarrow 0$, the value for $\bar{N} = \frac{1}{2}$ falls to the exact bound 1.39 of Proposition 2(a). Here and in Fig. 1, counts are on the whitened scale of Section 4.

Table 1: Guarantee on the lag-cancelled loop (2): a count at or below $\bar{N}$ guarantees an M_s no larger than the entry (exact).

	$\bar{N} = \frac{1}{2}$	$\bar{N} = 1$	$\bar{N} = \frac{3}{2}$
$\delta = 0.05$	1.615	2.058	2.562
$\delta = 0.02$	1.532	1.840	2.208
$\delta = 0.005$	1.473	1.668	1.918

2.3 Low counts can hide a high peak

Outside lag-cancelled loops, crossings and M_s come apart in both directions. Take a first-order plant with $T = 0.3L_d$ and PI with slow integral action, $T_i = 8T$, and K_c set for a gain margin of 1.62. The error never crosses the setpoint by more than 2%. The two counts that SPIN uses for PI stay below 0.1 turn. Yet $M_s = 2.62$. The oscillation rides on a slow decay that does not change sign. Only Γ_2 , which SPIN does not use for PI, sees it ($N_2 = 4.2$). Conversely, on $e^{-s}/(20s + 1)$ with PI ($K_c = 2.5$, $T_i = 5$), the error crosses twice with 26% overshoot, yet $M_s = 1.34$: the overshoot comes from the controller zero. In a grid of 630 first-order loops with $T/L_d \leq 5$, two or more crossings did mean $M_s \geq 1.69$, but this is a tendency, not a bound. So in general the counts can guide tuning but cannot certify it. Yet the same step still shows the peak (Fig. 2(a)); Section 3 explains why.

3 Result 2: The Same Step Gives M_s

Lemma 1 (One-step certificate) *Let the loop be linear, time invariant, stable and of one degree of freedom. Let e be the error after a unit setpoint step, and E its Fourier transform. Then*

$$S(j\omega) = j\omega E(j\omega) = 1 + \int_0^\infty \dot{e}(t) e^{-j\omega t} dt. \quad (4)$$

So $M_s = \max_\omega |S(j\omega)|$ follows from the record, and $L = 1/S - 1$ gives the gain and phase margins.

Proof: With $R = 1/s$ the transform of the step, $E = R/(1 + L) = RS$, so $S = sE$. The error jumps from 0 to 1 at $t = 0$. This jump gives the constant term. $\square$

We call the value read by (4) the certificate. Inside SPIN, it becomes the acceptance test of Section 4. The identity is exact. Any error in the reading comes from sampling, from noise, or from a record that has not settled. We computed the exact record of the loop (2) from its Lambert series. The peak read by (4) equals the true value $M_s = 2.23222$ to all five digits. On a zero-order-hold simulation of plant P_4 (Section 4) with SIMC PI, the reading error falls from 0.39% at 25 samples per dead time to 0.05% at 100 samples. A real controller is also sampled. The record then gives the M_s of the loop that actually runs. Fig. 2(b) shows the reading error on all loops of this paper. It is between -0.8% and $+0.8\%$ for the reference loops and for every tuning that certified SPIN accepted,

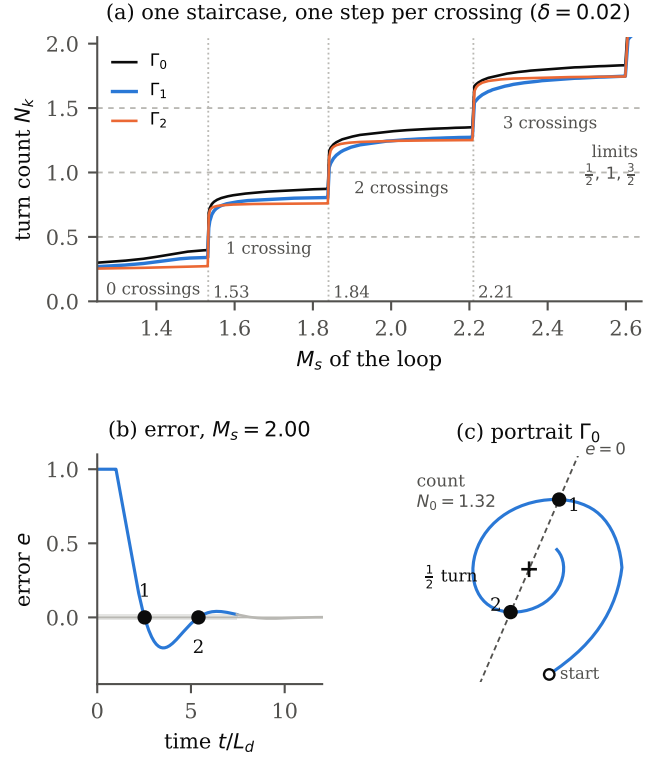


Figure 1: What a turn count is, on the lag-cancelled loop. (a) The three counts against M_s form one staircase, one step per setpoint crossing (steps at the values of Table 1); the limits $\frac{1}{2}, 1, \frac{3}{2}$ (dashed) lie in the gaps. (b) A record with $M_s = 2.00$ crosses twice before it settles within δ . (c) Its portrait Γ_0 : between the crossings (on the dashed line $e = 0$) the curve turns exactly half a turn around the end point (+). The radius is compressed for visibility; the angle is not.

and between -0.8% and $+1.3\%$ for the sharper peaks of the tunings that original SPIN accepted.

Corollary 1 (Two degrees of freedom) *Let the controller be $u = C_r r - C_{fb} y$, for example with setpoint weighting or with the derivative on the measurement. Let C_{fb} be known. Then, from the recorded u and y of one setpoint step,*

$$G(j\omega) = \frac{\int_0^\infty \dot{y} e^{-j\omega t} dt}{\int_0^\infty \dot{u} e^{-j\omega t} dt}, \quad S = \frac{1}{1 + C_{fb} G}, \quad (5)$$

where $\dot{u}$ includes the jump of u at $t = 0$.

Proof: $Y = GU$, so $sY/(sU) = G$. In a stable loop both derivatives decay exponentially, so their transforms exist. The loop transfer function is $C_{fb}G$. $\square$

We tested both forms on the plants of Section 4 with SIMC PID, sampled as in Section 4. For a one-degree-of-freedom loop, both forms are within 0.5% of the true M_s . With setpoint weight $b = 0$ and the derivative on the measurement, the simple form $j\omega E$ is wrong by -14% to $+8\%$, while form

(5) stays within 0.2% on all four plants. Noise biases the ratio of two noisy transforms in (5); our noise test below covers only the simple form.

Noise White noise on the record biases the reading up, to the safe side. Over 20 records on each of the four plants, 1% noise gave a bias of +0.1 to +0.6% (spread <0.6%), and 5% noise +1.3 to +4.9% (spread <2.5%).

Conditions The record must settle; otherwise the tail must be treated as in [7]. The sampling must be fast enough for the crossover frequency and for any derivative filter. The step must stay in the linear range, without actuator saturation.

4 Certified SPIN

The limits are half-turn limits After whitening each portrait with its second-moment matrix, a single oscillation gives three equal counts on the half turns of Section 2 (Fig. 1), and the limits $\bar{N} = (\frac{1}{2}, 1, \frac{3}{2})$ lie in the gaps. The per-axis limits (0.5, 0.75, 1.0) of SPIN separate the same 0|1, 1|2 and 2|3 crossings: on the lag-cancelled loop the per-axis counts lie in [0.06, 0.25], [0.55, 0.74] and [1.03, 1.24], those of Γ_2 half a turn lower. The limits make the count decisions robust, not the tuning.

Acceptance test When no count is above its limit, SPIN would raise all gains (line 7 of Algorithm 1). Algorithm 2 first reads $\hat{M}_s$ and its frequency ω_p from the same record by Lemma 1. If $\hat{M}_s \leq M^*$, the tuning is accepted. If not, the same record also tells which gain to lower, because it gives the loop $L = 1/S - 1$ and so the sensitivity of any other tuning of the same plant.

Proposition 3 (Shaping from one step) *Let C be a PID controller with gains K_k ($k = i, p, d$, the bands $\Gamma_0, \Gamma_1, \Gamma_2$; the derivative filter is tied to K_d/K_p), let S be read from the step, and let ω_p be the unique peak of $|S|$, $M = |S(j\omega_p)|$, $T = 1 - S$, $c_k = K_k \partial C / \partial K_k$.*

- (a) $g_k = \partial \ln M / \partial \ln K_k = -\text{Re}(T c_k / C)$ at ω_p , and $\sum_k g_k = \text{Re} S(j\omega_p) - 1$.
- (b) Lowering all gains lowers M to first order iff $\text{Re} S(j\omega_p) > 1$, that is iff $|L + \frac{1}{2}| < \frac{1}{2}$ at ω_p .
- (c) SPIN's move for band k (lower K_k , raise the bands below, all by the same factor) lowers M to first order iff $g_k > \sum_{j < k} g_j$. If $\sum_k g_k > 0$, such a band always exists.
- (d) For any other controller C' the sensitivity is $S' = 1/(1 + LC'/C)$, exactly, with $L = 1/S - 1$.

Proof: (a) $\partial \ln S / \partial \ln K_k = -(c_k/C)L/(1+L) = -T c_k/C$ and $\ln M = \text{Re} \ln S$ at the peak; the c_k add up to C , so $\sum g_k = -\text{Re} T$. (b) is $\sum g_k > 0$; and $\text{Re} \frac{1}{1+L} > 1 \Leftrightarrow 1 + \text{Re} L >$

$|1 + L|^2 \Leftrightarrow |L + \frac{1}{2}|^2 < \frac{1}{4}$. (c) The move changes $\ln M$ by $\ln \frac{1}{1-\beta} (\sum_{j < k} g_j - g_k)$. Let $P_k = \sum_{j \leq k} g_j$. The condition is $P_k > 2P_{k-1}$. If $\sum_k g_k > 0$, take the smallest k with $P_k > 0$: then $P_{k-1} \leq 0$ and the condition holds. (d) $L' = LC'/C$ for the same plant. $\square$

Three cautions apply. The first-order sign in (a)–(c) is wrong for about 1–2% of finite 10% steps, so Algorithm 2 uses (d), exact up to the reading error amplified by T'/T . $\text{Re} S > 1$ fails for about 1% of loops with strong integral action. And (d) predicts a frequency response, not stability; the next step checks it. On 934 moves of our runs with $\text{Re} S > 1$ and a unique peak, the condition held every time, the prediction was within 1.2% of the true M_s , and the true M_s fell in 99–100% of them.

Any band predicted to help reaches the target; we take the highest because it left the lower bands, which carry the load rejection, largest (see below). In Fig. 3(b), P_3 with PID, the step of the tuning that original SPIN accepts reads $\hat{M}_s = 1.81$ with $g = (0.44, 0.41, -0.05)$; lowering K_i is predicted to give 1.739 and does give 1.742.

Test We compare original SPIN, exactly as published, with certified SPIN, which adds only Algorithm 2 with $M^* = 1.7$. Both run for 200 iterations, filter the derivative with ratio 10, and keep each gain within $[0.001, 10]$ times its start value [1]. Because this box ties the result to the start, we run every loop from two starts: SIMC on a half-rule model with $\tau_c = 3\theta$, and a start twice as bold ($\tau_c = \theta$). The plants are the four plants of [1],

$$\begin{aligned} P_1 &= \frac{e^{-s}}{(10s+1)(s+1)^3}, & P_2 &= \frac{1.25 e^{-8s}}{(5s+1)^4}, \\ P_3 &= \frac{e^{-10s}}{(2s+1)(s+1)^3}, & P_4 &= \frac{e^{-4s}}{(8s+1)^6}, \end{aligned}$$

with PI and PID. Four more plants place the oscillation in a chosen band: a delay-dominant $e^{-5s}/(s+1)$ and two lag-dominant plants, $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$, with PI, and a fast plant $e^{-0.1s}/((s+1)(0.5s+1))$ with PID. We also use the family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d \in \{0.25, 0.5, 1, 2, 4, 8, 16\}$, so that the dead-time share $L_d/(L_d + \sum T)$ runs from 0.125 to 0.90. At the end, SPIN moves in a cycle between a few tunings. We report the most aggressive tuning that each rule accepts in its last 60 iterations. The controller is sampled at 1/20 of the derivative filter time constant (PID) or 1/1000 of the plant's time scale (PI).

Where the oscillation is decides Original SPIN ended above the target on 9 of the 19 loops from the cautious start (up to $M_s = 2.00$) and on 8 from the bold start (up to 2.07). In most PID runs the derivative gain ended at the ceiling of the box, so the box set the result and which loops failed depended on the start (P_1 and P_2 with PID passed from the cautious start and failed from the bold one; the lag-dominant PI loop did the

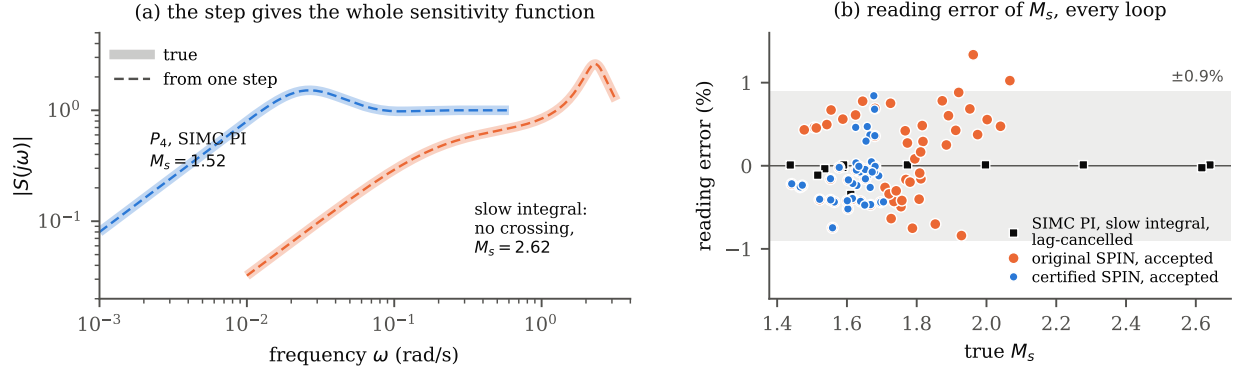


Figure 2: Lemma 1. (a) $|S|$ from one setpoint step (dashed) lies on the true one (thick), for P_4 with SIMC PI and for the slow-integral loop whose peak the counts miss. (b) Reading error of all 122 readings: 11 reference loops (SIMC PI on P_1 – P_4 , the slow-integral loop, the lag-cancelled loop for $k = 0.4, \dots, 0.9$) and the 111 distinct tunings accepted by original (69) or certified (42) SPIN in Section 4; the band is $\pm 0.9\%$.

Algorithm 2: certified SPIN: line 7 of Algorithm 1 becomes

- 7a: read $\hat{M}_s$, ω_p and $L = 1/S - 1$ from the same record
- 7b: **if** $\hat{M}_s \leq M^*$ **then** raise all gains by one step (accept)
- 7c: **else for each** band k : predict M_s after SPIN's move for band k by Prop. 3(d)
- 7d: apply the *highest* band whose predicted M_s is below $\hat{M}_s$; if none, lower all gains; continue at line 10

Default $M^* = 1.7$. Every accepted tuning then has $M_s \leq M^*$, up to the reading error.

opposite). Every failure was owned by Γ_1 or Γ_2 . The three loops owned by Γ_0 (delay-dominant PI, P_3 with PI, and the family at $L_d = 16$) stayed below the target from both starts; there the counts alone suffice. Certified SPIN ended at or below the target on all 19 loops from both starts. The highest true value was $M_s = 1.705$, read as 1.697. On the plant family (Fig. 3(c)), original SPIN exceeded the target at every dead-time share up to 0.82 (M_s from 1.71 to 1.81; from the bold start up to 2.07 at $L_d = 1$), and certified SPIN stayed between 1.58 and 1.68.

Which part does the work Without the box, the counts alone Original SPIN then went above the target on all 11 PID loops that converged (all but $L_d = 16$), up to $M_s = 2.04$. Certified SPIN ended at or below 1.705 on all of them (read 1.698 at most), so the target does not rely on any gain bound.

The price A lower peak costs load rejection. For a unit load step at the plant input, the integrated absolute error (IAE) of the cautious start rose by 0 to 15% on six of the nine loops whose tuning Algorithm 2 changed, by 24% on two, and by 52% on the lag-dominant PI loop; the fast PID loop, which original SPIN tuned to $M_s = 1.87$, lost nothing. The mean over all 19 loops was +7.7% (bold start: +5.1%). Choosing the band by the largest term of $|C|$ instead cost +11.1%, and

Table 2: True M_s of the most aggressive tuning that original and certified SPIN accept, from the cautious (3θ) and the bold (θ) start; the band that owns the peak of the original tuning from the cautious start; and the change of load-step IAE there.

		original		certified		owner	Δ IAE
		3θ	θ	3θ	θ		
P_1	PI	1.47	1.47	1.47	1.47	Γ_1	–
	PID	1.67	1.85	1.67	1.62	Γ_1	–
P_2	PI	1.64	1.67	1.64	1.67	Γ_1	–
	PID	1.60	1.78	1.60	1.66	Γ_1	–
P_3	PI	1.67	1.63	1.67	1.63	Γ_0	–
	PID	1.81	1.74	1.69	1.67	Γ_1	+24%
P_4	PI	1.55	1.60	1.55	1.60	Γ_1	–
	PID	1.62	1.56	1.62	1.56	Γ_1	–
delay-dom.	PI	1.68	1.63	1.68	1.63	Γ_0	–
lag-dom.	PI	2.00	1.68	1.67	1.68	Γ_1	+52%
2 lags	PI	1.44	1.47	1.44	1.47	Γ_1	–
fast	PID	1.87	1.97	1.66	1.68	Γ_2	0%

+88% on the fast loop.

5 Use in Practice

By eye, on a lag-cancelled loop: count how often the error crosses the setpoint before it stays within 2%. A SIMC loop crosses once; with $\lambda \geq (e - 1)L_d$ lambda tuning never crosses (Proposition 2(a)). *By computer*: every setpoint change in a historian gives M_s and the margins by Lemma 1 or Corollary 1, without a model.

Scope: this paper stays in theory and simulation: 19 simulated loops, two starts, the conditions of Section 3. As in [1, Remark 2], the counts assume noise below about 2% of the step. Noise in the counts and real loops are the subject of a follow-up. Scripts that

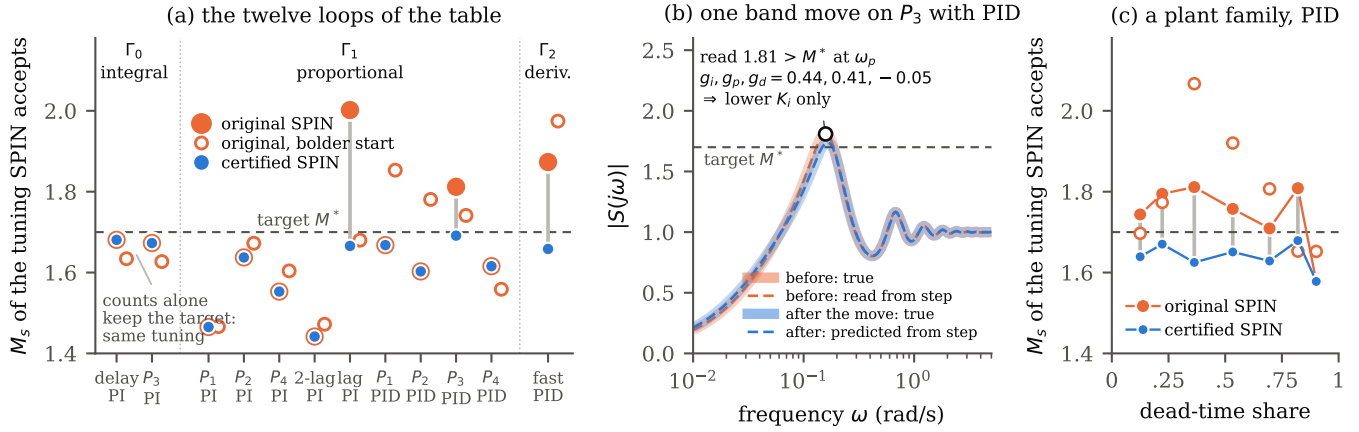


Figure 3: Original against certified SPIN ($M^* = 1.7$). (a) The loops of Table 2, grouped by the band that owns the peak of the original tuning. Filled orange: original, cautious start; open orange: original, bold start; blue: certified, cautious start. (b) One band move on P_3 with PID: the step reads $1.81 > M^*$ at ω_p ($\circ$), the gradient of Proposition 3 picks K_i , and the sensitivity predicted from the same step lies on the true one after the move. (c) The family $e^{-L_d s} / ((s+1)(0.5s+1)(0.25s+1))$ with PID against the dead-time share.

reproduce every number and figure are in the repository <https://github.com/ottapav/robopid-simulator>, folder docs/ECC27_certified_SPIN.

6 Conclusion

The results are twofold. On lag-cancelled loops a SPIN count has an exact meaning: half a turn per setpoint crossing, and the calibrated limits separate crossing numbers. On any other stable loop, counts and M_s are not tied, but the step that SPIN already records gives the peak and the loop, because its error is the step response of the sensitivity function. As an acceptance test with band shaping, it turned a start-dependent result (about half of our loops above the target) into a certified one (all loops at the target, within the reading error), at a mean cost of 5 to 8% in load rejection. A follow-up will study noise and real loops.

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